Research Assignment 1

Section A: Data Fundamentals

1. A Database is a digital repository for storing, managing and securing organized collections of data in real time.

Businesses use database analytics to spot trends, track inventory, and understand customer habits. Databases are effecient in handling big data

Real world examples

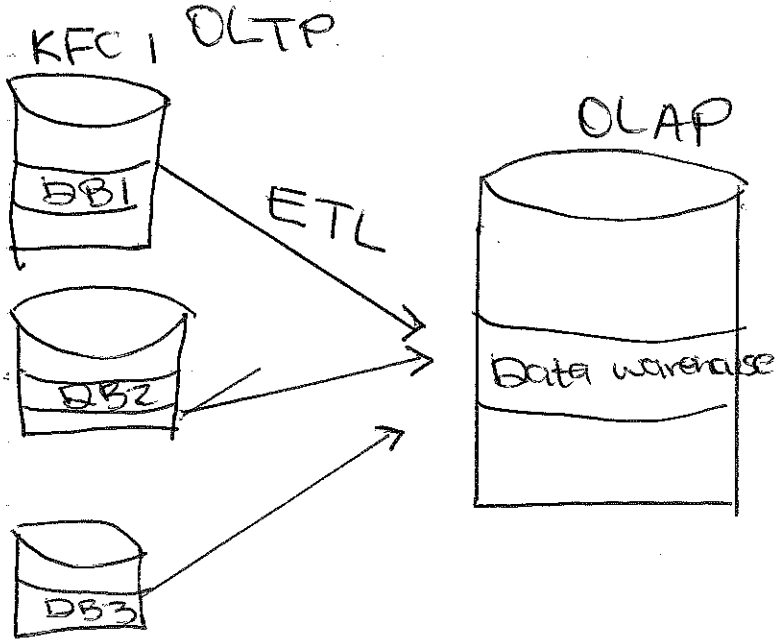
1. When you buy at KFC, your order details are being stored at an Online Transactional Processing Database (OLTP)
2. On social media, when people like specific content or shares it, the information is stored in a database and graphs can be used to analyse which content has alot of likes or shares.

2. Datawarehouse

Is a centralized digital storage system designed to collect, organize, and analyze large amounts of historical data from multiple sources. An Online Analytical Processing Database

It is different from a regular database because a regular database does not or is not used to analyze data, it is used to capture data.

Companies use ~~Data~~ warehouse to do data processing, data cleaning, reporting for better business decisions.



3. Data Lake

A data lake is a centalized storage repository that holds vast or extremely large amounts of raw, unprocessed data in its native format until it is needed.

Data warehouse store cleaned and processed data, whereas data lakes holds unproccessed data. Warehouses have built-in analytics engines and reporting tools, whereas data lake require external tools for processing.

Warehouse is best suitable for supporting the business intelligence and data analytics efforts of business users. Data lakes are best suited for operations that require large volumes of data in various data formats, such as artificial intelligence, machine learning and data science.

A business would prefer a data lake over a data warehouse when it needs to store massive amounts of raw, unstructured data e.g an airline or freight shipping company - collects millions of real-time weather logs, maintenance sensor redings and customer service audio recordings every hour.

4. Data Mart

A data mart is a subset of a data warehouse focused on a particular line of business, department or subject area.

A data warehouse stores intergrated, organization-wide data from multiple sources.
A data mart zoom in on a single subject area, function or team

Who typically uses it?
Department teams e.g marketing, sales, finance or human resources.

5.
Structured data (dimensions, tables and fact-tables) that can be connected to one another using primary key and foreign key.

~~D~~

Semi-structured data

It sits between structured and unstructured data formats. It does not follow a fixed schema but still includes semantic markers that define relationships and hierarchies.

Unstructured data

Information with no standardized format or data model, stored in its native form. Examples: emails, video, text documents, photographs.

6. Data Modeling

Is the process of creating a visual representation of either a whole information system or parts of it to communicate connections between data points and structures.

Data modeling set clear rules for data formats, reducing errors and improving overall data across teams. It helps engineers choose the right database technology and optimize storage for faster data retrieval.

It identifies design flaws, missing information, redundant structures early before writing code.
It acts as a common language between non-technical business stakeholders and technical developers

Two types of data models.

1. Relational Data Model
- It organizes data into tables with rows, columns, using primary and foreign keys to link related tables together efficiently for transactional applications

2. Dimensional Data Model
- Structuring data into fact tables (measurable events) and dimension tables (contextual attributes), optimized primarily for fast analytical reporting and business intelligence.

7. Data Mining

- Is the process of finding hidden patterns, correlations, and useful insights in large data sets using statistics, machine learning, and database systems.

Businesses use data mining to turn raw data into useful insights. This helps companies make smart choices, increase sales, and fix problems.

- Customer Segmentation: Grouping buyers by habits to create personal ads.

- Fraud detection: Spotting strange payment actions to stop theft righ away.

- Demand forecasting: Predicting future product sales to manage stock.

- Operating optimization: Finding slow steps in work tasks or supply chains

real-~~world~~ world examples

- Hyper-Personalized Retail Recommendations (e.g Amazon): Tracks what you view and buy. Data mining tools find patterns in these habits. The store then shows you personal item suggestions you are likely to buy.

- Real-Time Fraud Detection: Banks look at active payment actions. They compare new payments against your past history instantly. If a payment does not fit your normal pattern, the system blocks it as fraud.

8. DBMS - Database Management Systems

Is a software that enables users to create, store, retrieve, update, and manage complex data in a structured way.

four popular DBMS platforms:

1- MySQL - A widely used, open-source Relational Database Management Systems (RDBMS) that organizes data into traditional tables with rows and columns.

- Suitable Use case: E-commerce websites and Content Management Systems (CMS) like WordPress. Its structured nature is perfect for handling user accounts, product inventories, and transactional checkouts where data relations are highly predictable.

2. PostgreSQL

- Is an advanced, open-source Object-Relations DBMS known for its strict standards compliance and high extensibility. It natively supports both traditional SQL querying and advanced data types like JSON or spatial/geographic data

- Suitable Use Case: Financial technology (FinTech) apps or Geographic Information Systems (GIS). Its strong data integrity, compliance capabilities, and

complex query performance makes it ideal for systems requiring exact precision and geographical mapping.

3. Oracle Database

A powerful, proprietary enterprise-grade multi-model RDBMS developed by Oracle. It is built specifically to offer massive scaling, high-performance computing, strong security features, and disaster recovery for heavy data loads.

- Suitable for large-scale banking systems and global enterprise resource planning (ERP). Large institutions rely on Oracle to handle millions of mission-critical transactions per second with zero tolerance for system downtime.

4. MongoDB

It is a leading NOSQL, document-oriented database that stores data in flexible, JSON-like-documents rather than rigid tables. Fields can vary from document to document because it is a schema-free, allowing horizontal scalability across multiple servers.

Suitable for Real-time analytics, mobile apps, and Content Personalization Engines. It is ideal for rapidly changing datasets, such as user profile feeds, social media interactions, or IoT logs where the incoming data structure is unstructured or unpredictable.

9. ETL (Extract, transform, load)

Is a data integration process that combines, cleans and organizes data from multiple sources. It then loads that transformed data into a data warehouse, target database or another destination

ETL helps enterprises to improve data quality by filtering out duplicated, corrupted, or incomplete enteries, ensuring that analytics are based on accurate information.

It Unifies ~~Sei~~ Siloed data by merging data from disconnected departments and disparate apps into a single, cohesive view

It boost query performance and ensures compliance and security

Extract: Data is gathered or pulled from various original sources - such as databases, CRM software, APIs or cloud applications.

Transform: Business rules and logic are applied to process the raw data. Data is being cleaned (removing errors and duplicates), filtering unwanted fields, formatting dates, currencies into standard layouts, and combining tables.

Load - ~~is~~ The final, clean, and structured data is written and saved into a target destination, such as a central data warehouse.
This can happen in scheduled batches (like ~~Overning~~ overnight) or via continuous streaming.

10. Data storage formats

1. CSV (Comma-Separated Values)
Is a simple, text-based, row-oriented tabular format where columns are separated by commas. It has no built-in schema or data types.

Use it when you are dealing with small datasets, quick data exchanges, or when end users need to open and edit data directly in spreadsheet tools like Excel. Avoid it for large or complex data

2. JSON (JavaScript Object Notation)
Is a lightweight, text-based key-value and array format that supports nested and hierarchical data structures.

Use JSON for web applications, API responses, and semi-structured event logs where nested hierarchies matter more than analytical query speed.
Avoid it for heavy analytical scanning.

3. Parquet

It is a binary, column-oriented storage format optimized for big data. It compresses data efficiently by column and stores embedded metad

Use it for large-scale analytical workloads (OLAP), Cloud data lakes, and querying specific columns across millions or billions of rows.
Avoid it for row-level writes or human readability

4. Avro

- A row-oriented binary serialization format that stores schemas inside JSON metadata and supports robust schema evolution.

Use for high-throughput data streaming pipelines like Kafka or write-heavy ingestion where schemas change frequently over time.
Avoid it for read-heavy analytical reporting.

Section B: Artificial Intelligence & Machine Learning

11. Artificial Intelligence (AI)

Is the simulation of human intelligence by computer systems and machines.

Narrow AI (weak AI)
- Are systems designed to perform a single, dedicated, or narrow set of tasks within a specific boundary. It cannot function outside its

-Programmed scope. It is the only type of AI that exists and is used in the world today. Eg Apple Siri

Strong AI

- Also known as Artificial General Intelligence (AGI), it refers to machines that possess human-level cognitive abilities, meaning they can learn, reason, and apply knowledge across any ~~Intelligence~~ Intellectual task just like a human. Eg a hypothetical robot assistant that can drive a car, robot that can write a code.

12. Machine Learning.

-Is a subset of artificial intelligence that uses algorithms to find patterns in data and make predictions without needing hard-coded rules.

How it differs from Traditional Programming:
The main difference lies in how a computer gets its instructions to solve a problem.

Traditional Programming: You input Data(Input) + Rules(Logic) into the computer, and it outputs the answers. A human programmer must write every explicit condition and instruction.

Machine Learning: You input Data(Input) + answers (Output/examples) into the computer, and the algorithm figures out the Rules(Logic) on its own

by analysing patterns.

Three main types of Machine Learning

1. Supervised machine learning

Is a type of machine learning where the model is trained on a labelled dataset (the target or outcome is known)

- It is used commonly for risk assessment, image recognition, predictive analytics and fraud ~~dect~~ detection.

- Regression algorithms - It predicts output value by identifying linear relationships between real or continuous values (e.g temperature, salary)

2. Unsupervised machine learning

- Draw inferences from unlabeled datasets, facilitating exploratory data analysis and enabling pattern ~~reg~~ recognition and predictive modeling.

- Cluster analysis is a common unsupervised learning method, which uses clustering algorithms to categorize data points according to value similarity. Consists of: K-means clustering, Hierarchical clustering and probabilistic clustering.

3. Reinforcement Learning

Training an agent to make decisions through trial and error by interacting with a dynamic environment. The computer receives positive rewards for correct actions and negative penalties for wrong ones, learning over time to maximize its total reward

13. Deep Learning

Is a specialized subset of machine learning that uses multi-layered artificial neutral networks to automatically learn complex patterns with vast amount of data eg text & audio with minimum human intervention.

Artificial Intelligence (AI) is the broad, overarching science of creating machines that can mimic human intelligence to solve problems and perform tasks.

Neutral networks

- An Artificial neutral network (ANN) is a computational model inspired by the biological network of neurons in the human brain. It serves as a fundamental building block or backbone of deep learning.

They process information through a continuous cycle of prediction, error measurement and adjustment.